

Calibration results

=====

Normalized Residuals

Reprojection error (cam0): mean 0.7904524925415563, median 0.6345240730299633, std: 0.785984831361436
Gyroscope error (imu0): mean 3.8014817855144494, median 2.2609059501061015, std: 7.997100955314848
Accelerometer error (imu0): mean 8.664268138205964, median 5.203909794220246, std: 16.645481349581313

Residuals

Reprojection error (cam0) [px]: mean 0.7904524925415563, median 0.6345240730299633, std: 0.785984831361436
Gyroscope error (imu0) [rad/s]: mean 0.0021806914560028555, median 0.00129695170630817, std: 0.004587476860338904
Accelerometer error (imu0) [m/s^2]: mean 0.06138755228400049, median 0.036870429155492934, std: 0.11793556482098425

Transformation (cam0):

T_ci: (imu0 to cam0):

[[-0.99974616 0.02240266 -0.00239394 0.12704628]
[0.00299472 0.0268223 -0.99963573 0.01396459]
[-0.02233028 -0.99938916 -0.02688258 -0.0684405]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[-0.99974616 0.00299472 -0.02233028 0.12544392]
[0.02240266 0.0268223 -0.99938916 -0.07161943]
[-0.00239394 -0.99963573 -0.02688258 0.01242379]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.07873365764124837

Gravity vector in target coords: [m/s^2]

[0.17629187 1.74454791 -9.64851785]

Calibration configuration

=====

cam0

Camera model: pinhole

Focal length: [158.7706573659226, 159.24200551485708]

Principal point: [158.60457216458644, 158.95712838526592]

Distortion model: radtan

Distortion coefficients: [-0.006922802778356867, -0.000568858503475161, -0.0003298384944185784,
-0.0042518729236620326]

Type: checkerboard

Rows

Count: 8

Distance: 0.02 [m]

Cols

Count: 11

Distance: 0.02 [m]

IMU configuration

=====

IMU0:

Model: calibrated

Update rate: 50.0

Accelerometer:

Noise density: 0.0010019901002151698

Noise density (discrete): 0.007085139945439349

Random walk: 1.9893785362909353e-05

Gyroscope:

Noise density: 8.112529814510266e-05

Noise density (discrete): 0.0005736424844418254

Random walk: 6.28721982800146e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

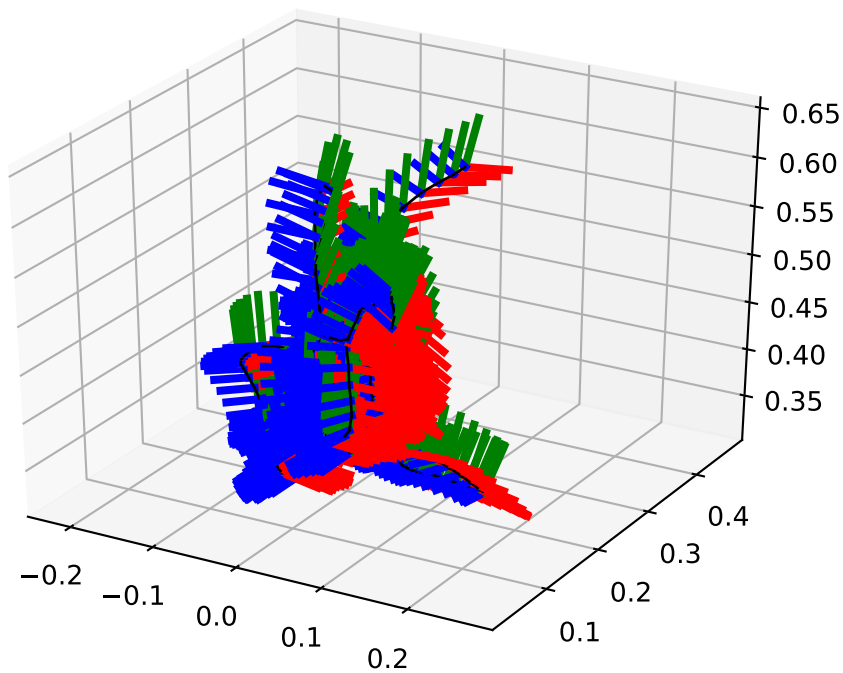
[0. 1. 0. 0.]

[0. 0. 1. 0.]

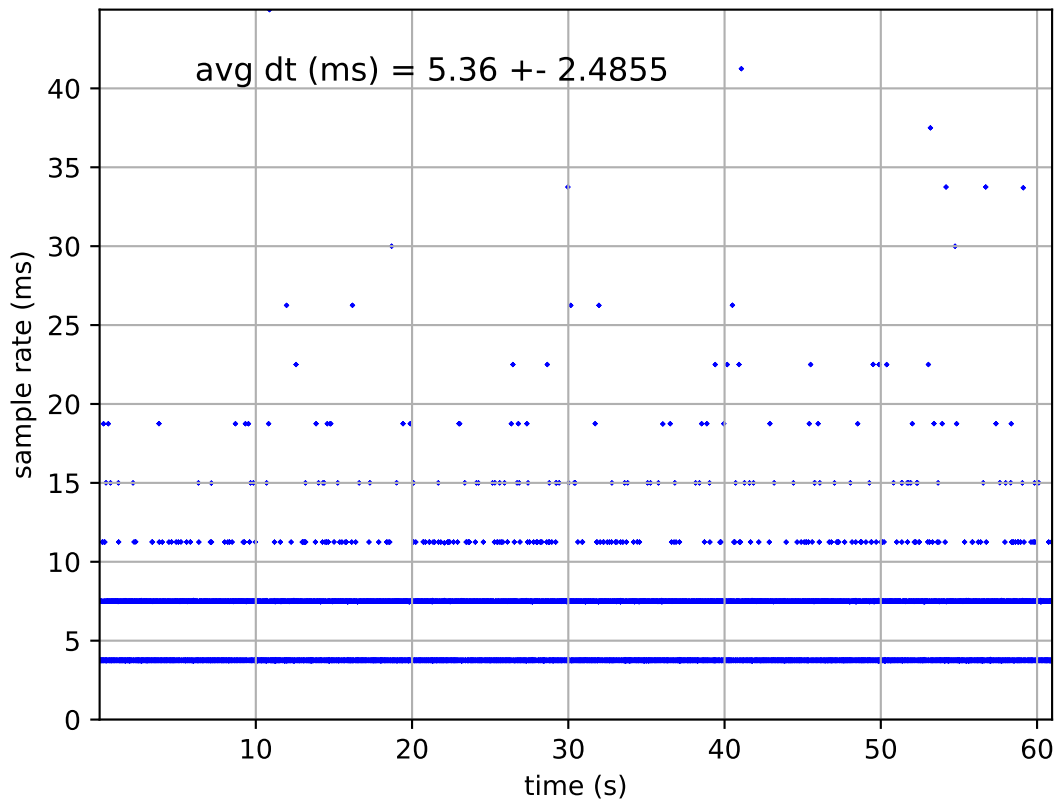
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

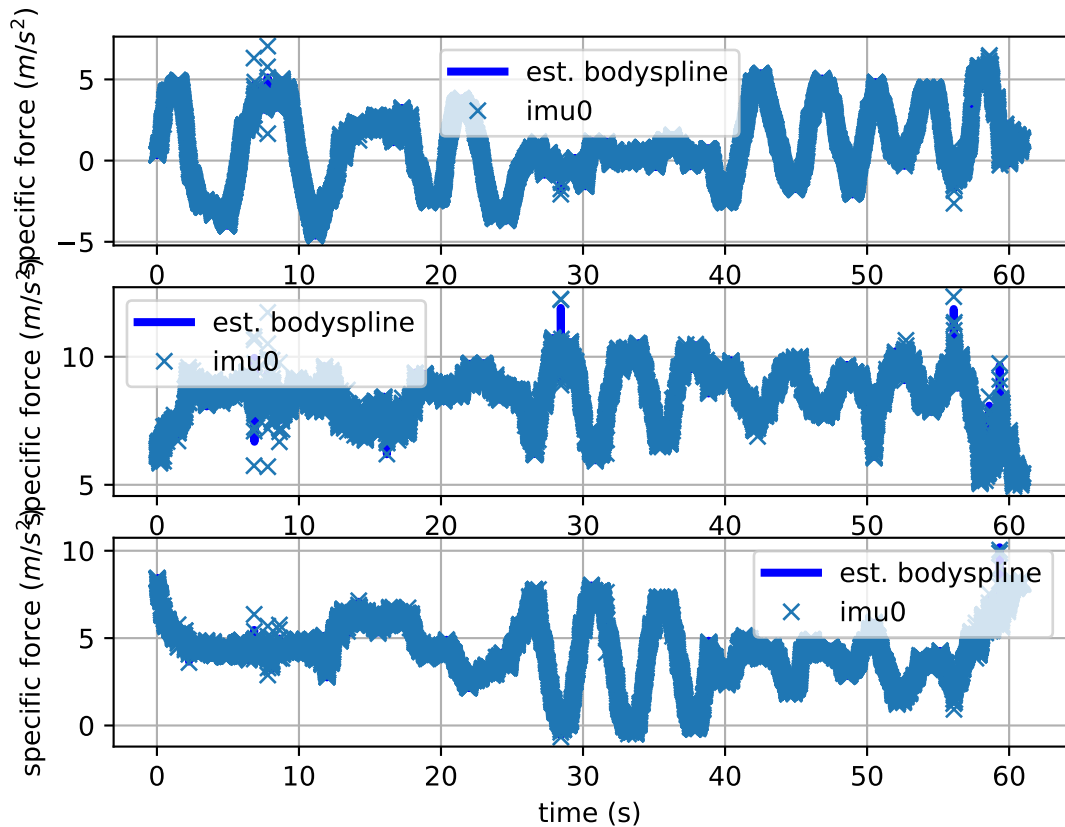
imu0: estimated poses



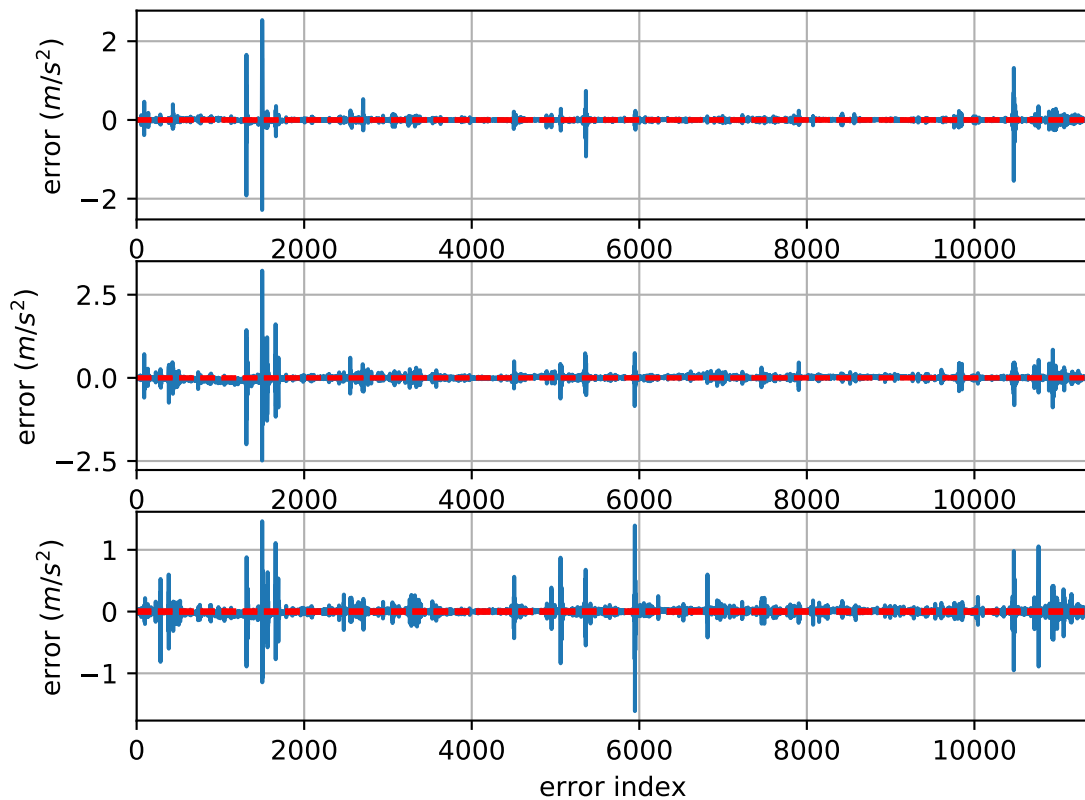
imu0: sample inertial rate



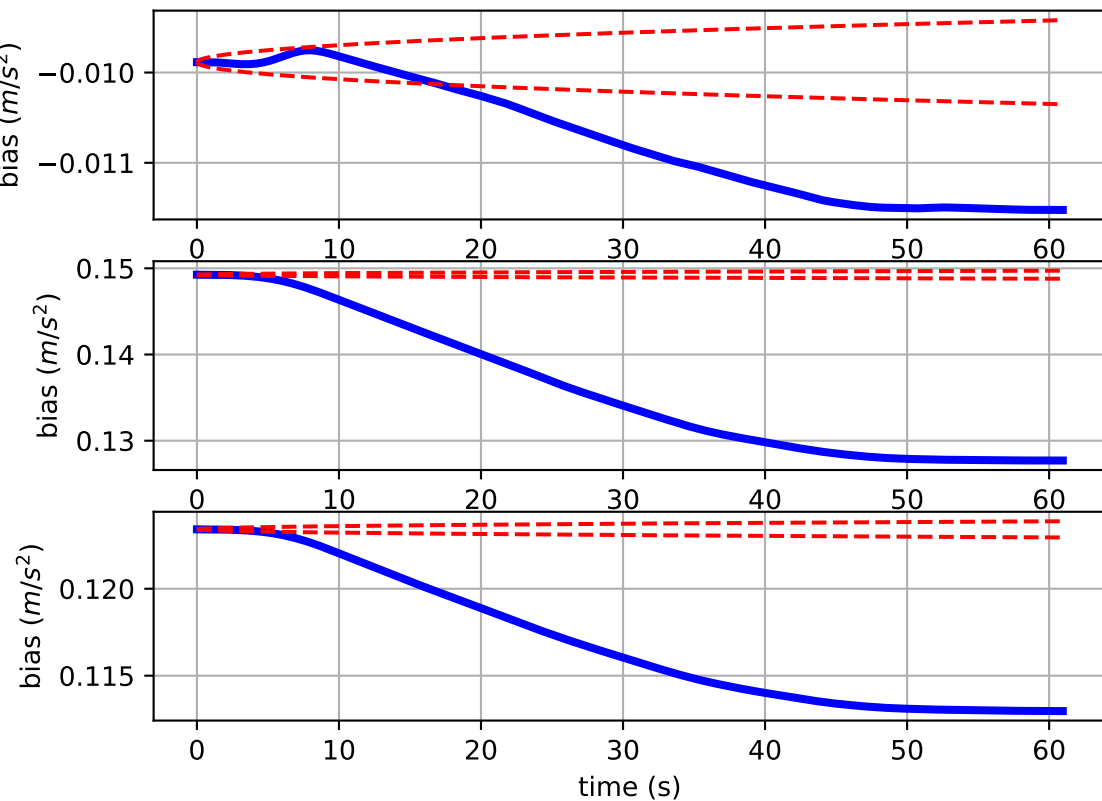
Comparison of predicted and measured specific force (imu0 frame)



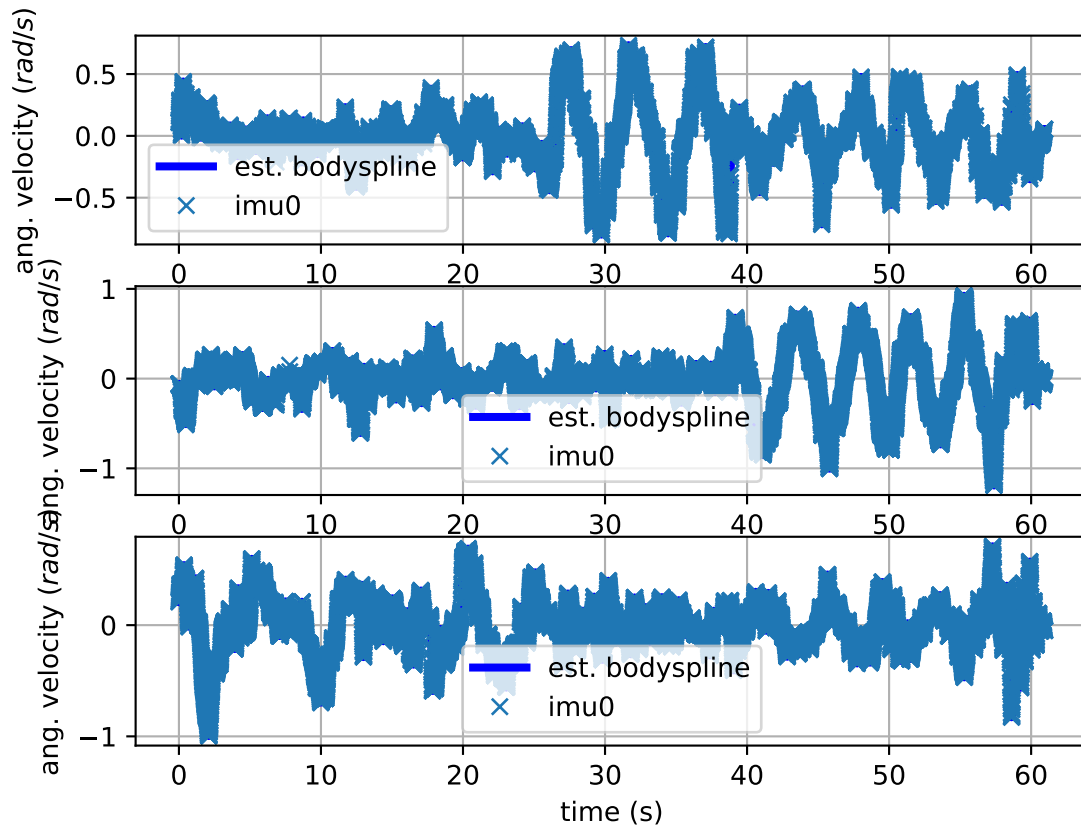
imu0: acceleration error



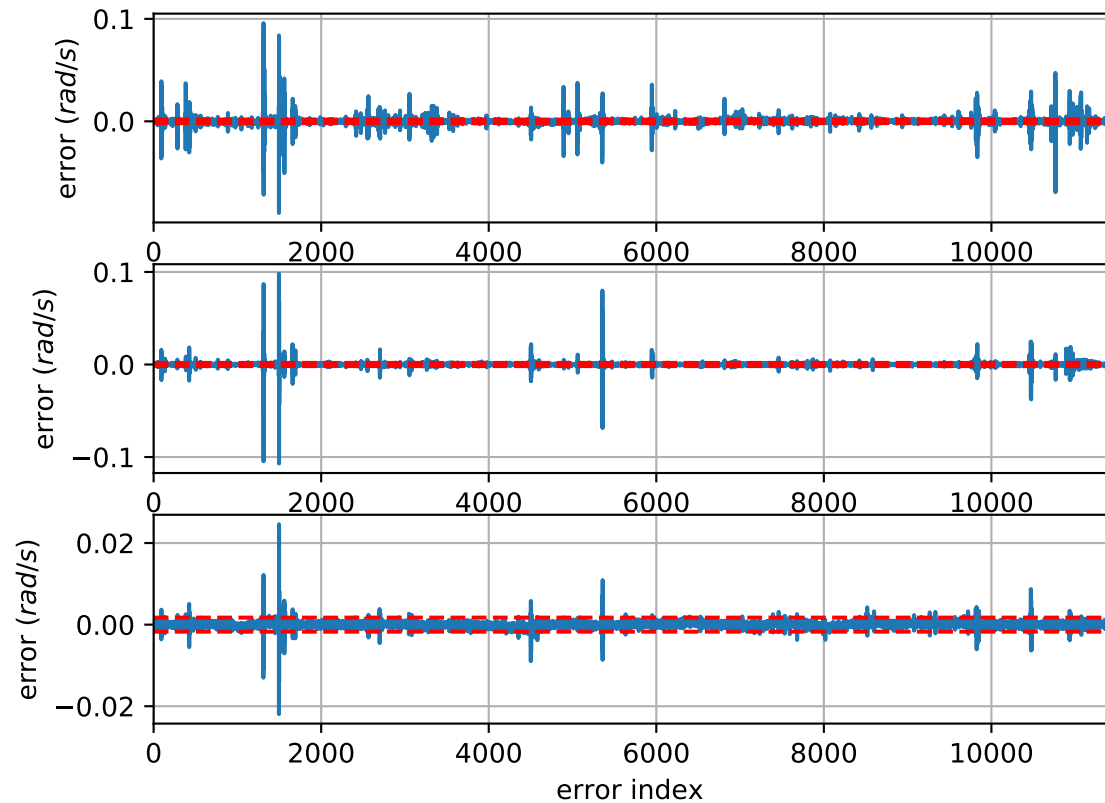
imu0: estimated accelerometer bias (imu frame)



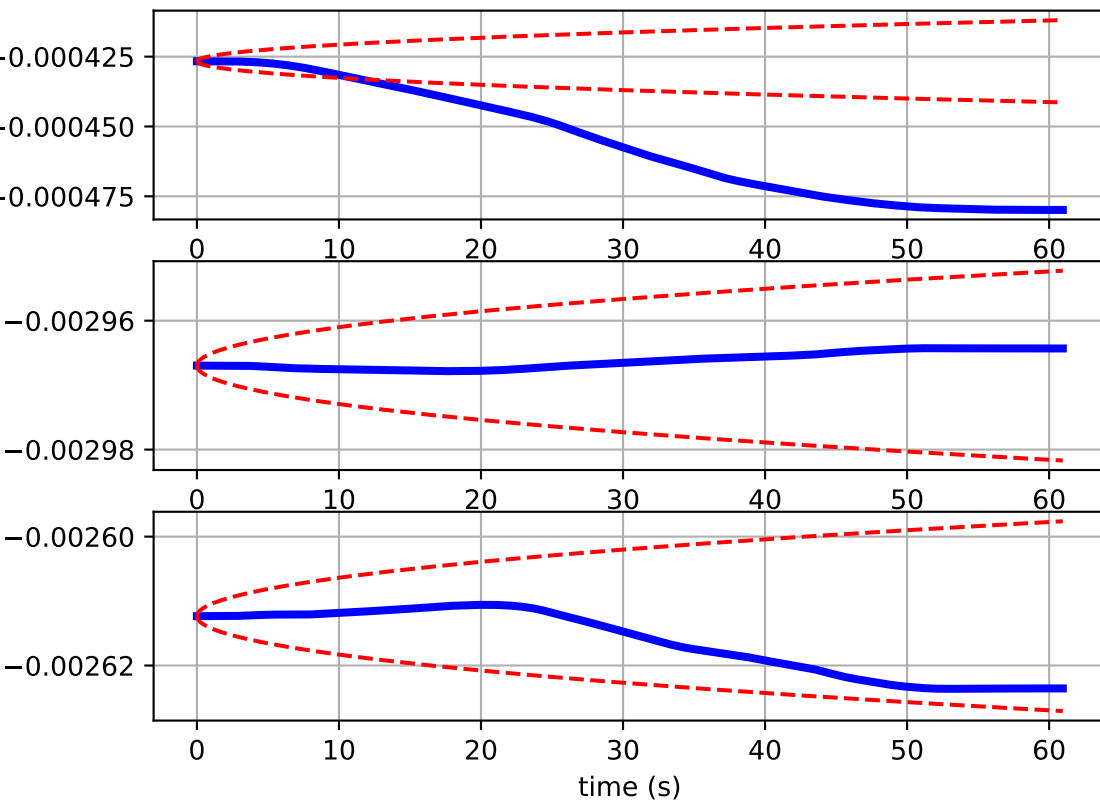
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

